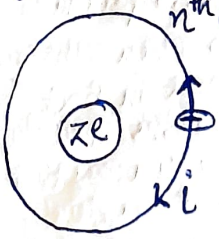
Magnetic Properties of matter

Magnetic moment of H-like atom:



$$\vec{M} = \frac{Q}{2m}\vec{L}$$

$$\boxed{M = \frac{e}{2m}\frac{nh}{2\pi}}$$ orbital magnetic moment

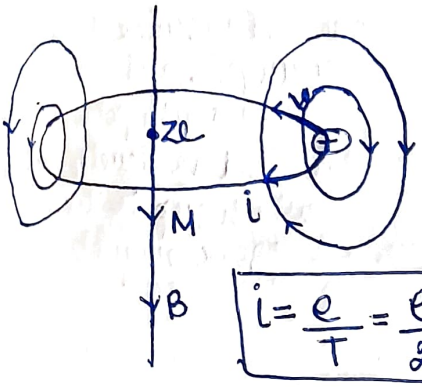
For H-atom, n=1 (ground state)

$$\Rightarrow \boxed{M_0 = \frac{eh}{4m\pi}}$$ Bohr magneton

$n=2 \quad M = 2M_0$

$n=3 \quad M = 3M_0$

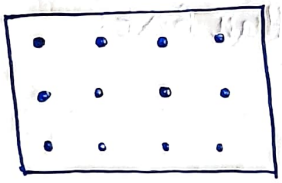
$n=n \quad M = nM_0$



$$\boxed{i = \frac{e}{T} = \frac{ev}{2\pi r}}$$

→ Apart from orbital motion of $e^{\ominus}$, there is magnetic dipole moment due to spin of e- also: $M_s = 9.285 \times 10^{-24} \frac{J}{T}$

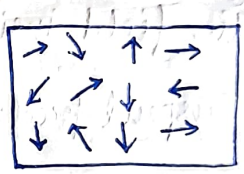
##



$M_{atom} = 0$

$M_{total} = 0$

$B = 0$



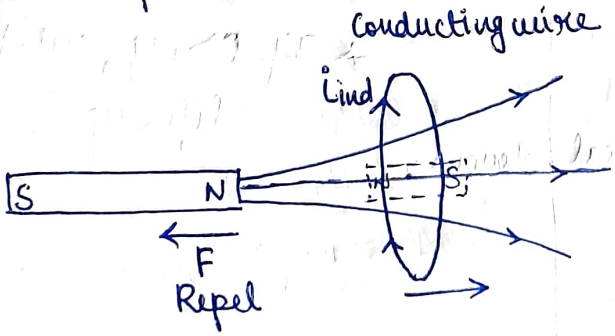
Random direction

$M_{atom} \neq 0$

$M_{total} = 0$

$B = 0$

Faradays law of induction:



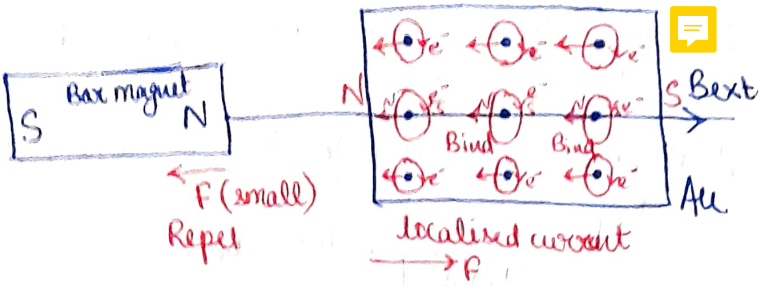
Diamagnetic Substance:

→ Each atom has zero magnetic dipole moment

→ Cu, Zn, Bi, Ag, Au, NaCl, H_2O

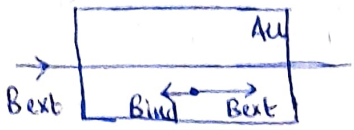
Diamagnetism (Application of Faraday's law):

→ It is a weak phenomenon



→ Acc. to faraday's law when substance is kept in Bext, $e^{\ominus}$ in atoms start revolving in such a way that a current is developed which opposes Bext. N-S, cancel and finally one end behaves like north pole and other like south pole. A Bind is developed which repels the bar magnet.

→ Diamagnetic substances repel bar magnet and repel ext. magnetic field.



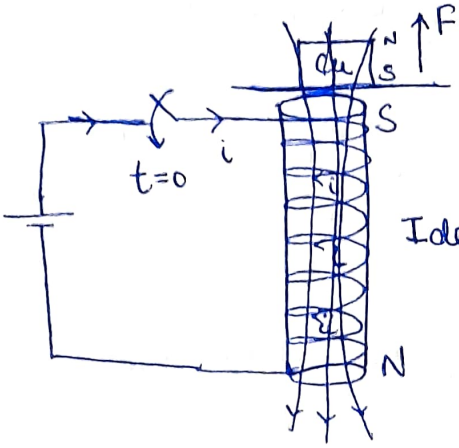
$B_{net} = B_{ext} - B_{ind}$

$B_{net} < B_{ext}$

→ If a diamagnetic substance is placed inside ext. magnetic field, then magnetic field inside it will be less than Bext.

Diamagnetic substances repel bar magnet and repel external magnetic field

Levitation: Suspend freely in air



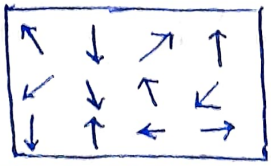
* If F > mg, substance lifts up.

this is levitation

Paramagnetic Substances:

→ Each atom has some magnetic dipole moment.

→ FeO, O_2, Titanium, Al

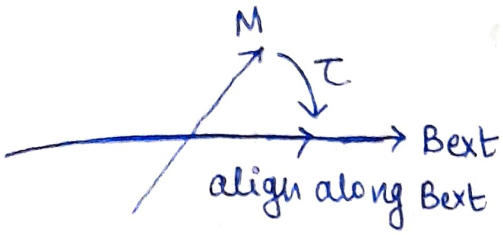


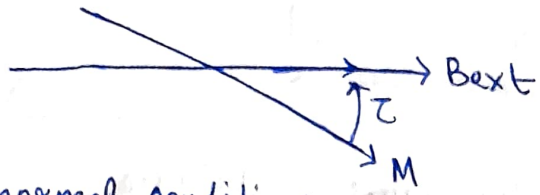
$B_{ext} = 0$

$M_{atom} \neq 0$

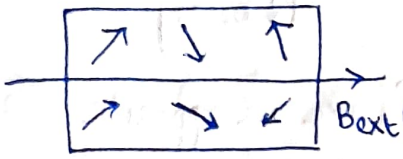
$M_{Al} = 0$

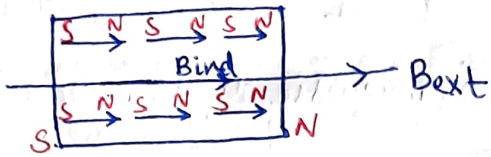
Paramagnetism:



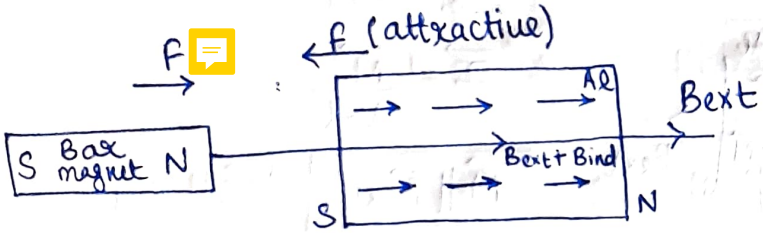


→ There is partial alignment in normal conditions.
→ As Bext ↑, alignment ↑
→ As temp ↑, KE_{atom} ↑, alignment ↓
→ For proper alignment, Bext ↑, temp ↓.





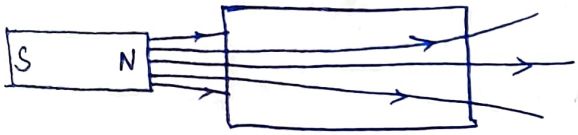
At very high value of Bext, all the magnetic dipoles will align along $\vec{B}_{ext}$ and this situation is known as magnetically saturated.



$B_{net} = B_{ext} + B_{ind}$

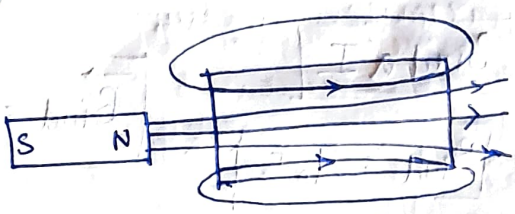
$B_{net} > B_{ext}$

→ Paramagnetic substances attract magnetic field and bar magnetic.



Cu (Diamagnetic)

As $B_{net} < B_{ext}$, MLOF: outsite ⇒ more
inside ⇒ less



Titanium (Paramagnetic)

As $B_{net} > B_{ext}$, MLOF: inside ⇒ more
outside ⇒ less

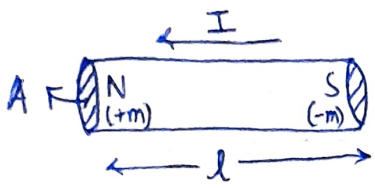
Intensity of Magnetization ($\vec{I}$):

→ It is defined as magnetic dipole moment per unit volume. **(magnetic dipole moment density)**

→ Vector Quantity
→ Direction: Along dipole moment ($\vec{M}$)
→ Unit = $\frac{Amp\text{-}m^2}{m^3} = \frac{Amp}{meter}$
→ Dimension = $[IL^{-1}]$

$$\vec{I} = \frac{\vec{M}}{V}$$

$\vec{I}$ of a bar magnet :

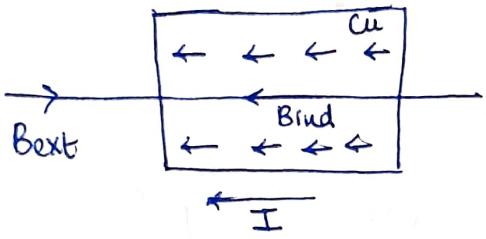


$$I = \frac{M}{V} = \frac{ml}{Al} = \frac{m}{A}$$

$$\boxed{I = \frac{m}{A}}$$

direction : S to N

$\vec{I}$ of a diamagnetic substance :



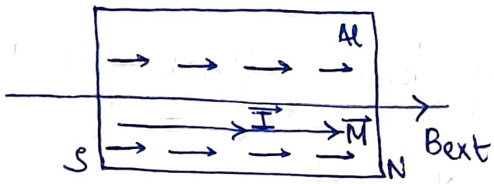
$$\Sigma \vec{M_i} = \vec{M}$$

$$\vec{I} = \frac{\vec{M}}{Vol.}$$

$\boxed{\vec{M} \uparrow\downarrow \vec{B}_{ext}}$ $\boxed{\vec{I} \uparrow\downarrow \vec{B}_{ext}}$

→ Induced $\vec{B}$, magnetic dipole moment and $\vec{I}$ are all opposite to $\vec{B}_{ext}$.

$\vec{I}$ of a paramagnetic substance :



$$\Sigma \vec{M_i} = \vec{M}$$

$$\vec{I} = \frac{\vec{M}}{Vol}$$

$\boxed{\vec{M} \parallel \vec{B}_{ext}}$ $\boxed{\vec{I} \parallel \vec{B}_{ext}}$

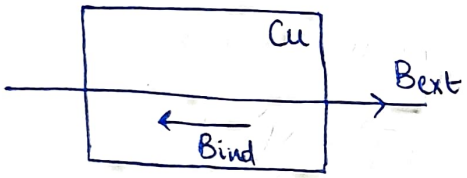
→ $\vec{B}_{ind}$, $\vec{M}$ and $\vec{I}$ are all along $\vec{B}_{ext}$.

* $\boxed{B_{ind} \propto I}$ $\boxed{B_{ind} = \mu_0 I}$

$\boxed{\vec{B}_{ind} = \mu_0 \vec{I}}$

for both diamagnetic and paramagnetic substances

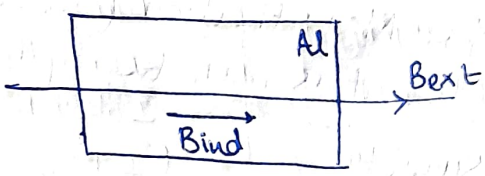
Case 1) Diamagnetic



$$B_{net} = B_{ext} - B_{ind}$$

$$\boxed{B_{net} = B_{ext} - \mu_0 I}$$

Case 2) Paramagnetic



$$B_{net} = B_{ext} + B_{ind}$$

$$\boxed{B_{net} = B_{ext} + \mu_0 I}$$

Magnetic Intensity (H):

We know, magnetic induction/field ($\vec{B}$) is medium dependent.
For diamagnetic medium, B↓ whereas for paramagnetic medium, B↑.
So, we define magnetic intensity (H), which doesn't depend on medium.

In air, $\vec{H} = \frac{\vec{B_0}}{\mu_0}$ ($\vec{B_0}$ = magnetic induction in vacuum/air)

$\vec{B_0} = \mu_0 \vec{H}$

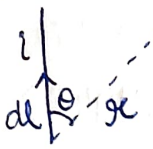
In any other medium,

$\vec{H} = \frac{\vec{B}}{\mu}$ ($\vec{B}$ = magnetic induction in medium)

$\vec{B} = \mu \vec{H}$ (μ = permeability of medium)

Ex ① Biot-Savart law:

air



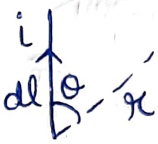
$B_0 = \frac{\mu_0}{4\pi} \frac{i\,dl \sin\theta}{r^2}$

$H = \frac{B_0}{\mu_0} = \frac{i\,dl\sin\theta}{4\pi r^2}$

thus, H same in both mediums

medium

Given: μ_r = relative permeability



$B = \frac{\mu}{4\pi} \frac{i\,dl\sin\theta}{r^2}$

$H = \frac{B}{\mu} = \frac{i\,dl\sin\theta}{4\pi r^2}$

We know, $\mu_r = \frac{\mu}{\mu_0} \Rightarrow \mu = \mu_0 \mu_r$

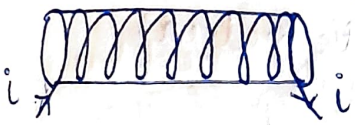
$B = \frac{\mu_0 \mu_r}{4\pi} \frac{i\,dl\sin\theta}{r^2}$

$\therefore \boxed{B = \mu_r B_0}$

② Ideal Solenoid:

air

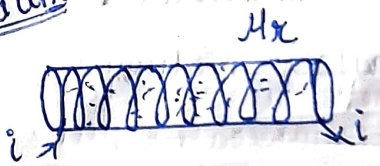
$n = \frac{\text{no. of turns}}{\text{length}}$



$B_0 = \mu_0 n i$

$H = \frac{B_0}{\mu_0} = ni$

medium



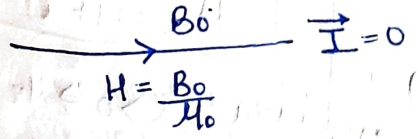
$B = \mu n i = \mu_0 \mu_r n i = \mu_r B_0$

$H = \frac{B}{\mu} = \frac{B}{\mu_0 \mu_r} = ni$

Relation b/w $\vec{B}$ and $\vec{H}$:

Case I: Vacuum

In vacuum, there are no atoms ⇒ magnetization = 0
⇒ $B_{ind} = 0 \Rightarrow \vec{I} = 0$



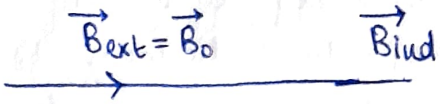
$\vec{B} = \vec{B_0} + \vec{B_{ind}} = \vec{B_0} + \mu_0 \vec{I}^{\,0}$

$\star \boxed{\vec{B} = \vec{B_0}}$

$\star \boxed{\vec{H} = \frac{\vec{B_0}}{\mu_0}}$

$\boxed{\vec{B_0} = \mu_0 \vec{H}}$

Case 2: Any other medium



$$\vec{B} = \vec{B}_{ext} + \vec{B}_{ind}$$
$$\vec{B} = \vec{B}_0 + \mu_0 \vec{I}$$
$$\vec{B} = \mu_0 \vec{H} + \mu_0 \vec{I}$$
$$\star \boxed{\vec{B} = \mu_0(\vec{H} + \vec{I})}$$

Now, $\vec{B} = \mu_0(\vec{H} + \vec{I})$

$= \mu \vec{H} = \mu_0(\vec{H} + \vec{I})$

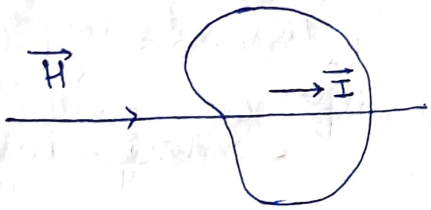
$= \mu_0 \mu_r \vec{H} = \mu_0(\vec{H} + \vec{I})$

$$\star \boxed{\mu_r \vec{H} = \vec{H} + \vec{I}}$$

→ For Paramagnetic: $\vec{I} \uparrow\uparrow \vec{H} \Rightarrow B > \mu_0 H$

→ For Diamagnetic: $\vec{I} \uparrow\downarrow \vec{H} \Rightarrow B < \mu_0 H$

Magnetic Susceptibility (χ):



$$\star \boxed{\chi = \frac{I}{H}}$$

→ less H, more magnetization, High I ⇒ $\chi \uparrow$

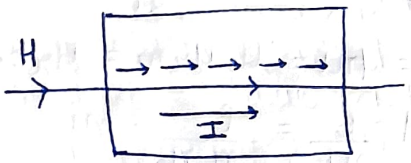
→ more H, less magnetization, low I ⇒ $\chi \downarrow$

→ High magnetic susceptibility means, easy to magnetize.

→ In Vacuum, $\vec{I} = 0 \Rightarrow \chi = 0$

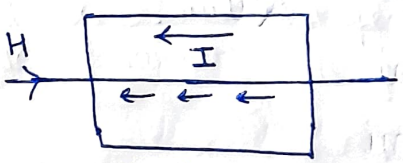
Now, $\star \boxed{\vec{I} = \chi \vec{H}}$

① Paramagnetic medium:



As $\vec{I} \uparrow\uparrow \vec{H} \Rightarrow \chi = +ve$

② Diamagnetic medium:



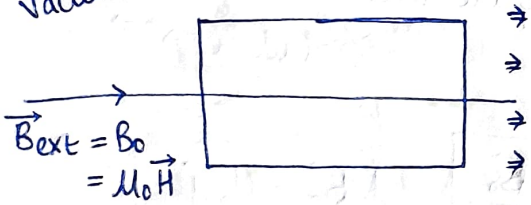
As $\vec{I} \uparrow\downarrow \vec{H} \Rightarrow \chi = -ve$

Para: Al $\chi = 2.1 \times 10^{-5}$
Tungsten $\chi = 6.8 \times 10^{-5}$

Dia: Cu = -0.96×10^{-5}
Ag = -2.6×10^{-5}
Au = -3.6×10^{-5}
Bi = -1.66×10^{-4}

Now,

Vacuum



$\vec{B} = \vec{B}_{ext} + \vec{B}_{ind}$

$\Rightarrow \vec{B} = \vec{B}_0 + \vec{B}_{ind}$

$\Rightarrow \vec{B} = \mu_0 \vec{H} + \mu_0 \vec{I}$

$\Rightarrow \vec{B} = \mu_0(\vec{H} + \vec{I})$

$\Rightarrow \vec{B} = \mu_0(\vec{H} + \chi \vec{H})$

$\Rightarrow \vec{B} = \mu_0 \vec{H}(1 + \chi)$

$\Rightarrow \mu \vec{H} = \mu_0 \vec{H}(1 + \chi)$

$\Rightarrow \mu = \mu_0(1 + \chi)$

$\rightarrow \mu_0 \mu_r = \mu_0(1 + \chi)$

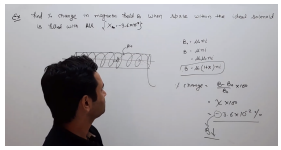
$\therefore$ Relative Permeability $\boxed{\mu_r = (1 + \chi)}$

Para: $\chi > 0 \Rightarrow \mu_r > 1$

Dia: $\chi < 0 \Rightarrow \mu_r < 1$

Vaccum: $\chi = 0 \Rightarrow \mu_r = 1$

Q Find % change in magnetic field B when space within the ideal solenoid is filled with Al. ($\chi_{Al} = 2.1 \times 10^{-5}$)



$B_0 = \mu_0 ni$

When it is filled with Al,

$B = \mu ni = \mu_0 \mu_r ni$

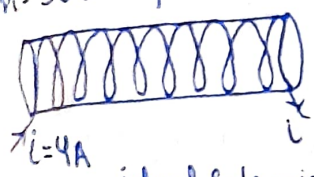
$B = \mu_0 (1+\chi) ni$

$$\% \text{ change} = \frac{B - B_0}{B_0} \times 100$$

$$= \frac{\mu_0 ni(1+\chi) - \mu_0 ni}{\mu_0 ni} \times 100$$

$$= \chi \times 100 = 2.1 \times 10^{-5} \times 100 = \underbrace{2.1 \times 10^{-3}\%}_{+\text{ve value} \Rightarrow B\uparrow}$$

Q $n = 50$ turns/cm



ideal solenoid.

Find ① Magnetic field B

$$B = \mu_0 ni = 4\pi \times 10^{-7} \times 5000 \times 4$$

$$= 8\pi \times 10^{-3}$$

$$= 8 \times 3.14 \times 10^{-3} \text{ T}$$

$$\boxed{B = 25 \text{ mT}}$$

② Magnetic intensity H

$$H = \frac{B_0}{\mu_0} = \frac{\mu_0 ni}{\mu_0} = ni = 5000 \times 4$$

$$\boxed{H = 20000 \text{ Amp/m}}$$

③ An iron rod is inserted inside solenoid which creates $I = 4 \times 10^6$ A/m.
Find new values of (i) H (ii) B

(i) H doesn't change with medium

$\therefore H = 20000$ A/m

(ii) $\vec{B} = \vec{B}_0 + \vec{B}_{ind} = \mu_0 \vec{H} + \mu_0 \vec{I}$

$= 25\text{mT} + 4\pi \times 10^{-7} \times 4 \times 10^6$

$= 25\text{mT} + 16\pi \times 10^{-1}$ T

$= 25\text{mT} + 1.6 \times 3.14$

$= 0.025$ T $+ 5.024$

$= 5.049$ T

④ Find permeability of Fe.

$B = \mu H \Rightarrow \mu = \frac{B}{H}$

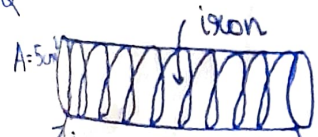
$\mu = \frac{5.049}{20000} = 2.52 \times 10^{-4}$

⑤ Find μ_r of Fe.

$\mu_r = \frac{\mu}{\mu_0} = \frac{2.52 \times 10^{-4}}{4\pi \times 10^{-7}} = 200.63$

⑥ Find χ $\Rightarrow \chi = \mu_r - 1 = 199.63$

Q



$n = 2000 \frac{\text{turns}}{\text{meter}}$

ideal solenoid

Given B inside the core is 1.57 T

Find ① Intensity of magnetization I

$B = B_0 + B_{ind} = \mu_0 \vec{H} + \mu_0 \vec{I}$

$$\vec{I} = \frac{B}{\mu_0} - H \qquad \left(H = \frac{B_0}{\mu_0} = ni\right)$$

$$= \frac{1.57}{4\pi \times 10^{-7}} - 4000$$

$I = 12.5 \times 10^5 - 4000 = 121000$ A/m

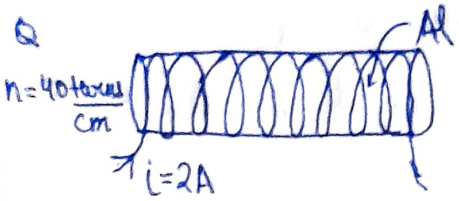
② Pole strength

$I = \frac{M}{V} = \frac{ml}{Al} = \frac{m}{A}$

$m = IA = 12.5 \times 10^5 \times 5 \times 10^{-4}$

$m = 625$ A-m

Q



ideal Solenoid

Given $\chi_{Al} = 2.3 \times 10^{-5}$

Find ① Magnetic Intensity, H

$$H = \frac{B_0}{\mu_0} = ni = 4000 \times 2 = 8000 \text{ A/m}$$

② Intensity of magnetization I

$$I = \chi H = 2.3 \times 10^{-5} \times 8000$$
$$= 18.4 \times 10^{-2} = 0.184 \text{ A/m}$$

③ Magnetic field B

$$B = B_0 + B_{ind} = \mu_0 H + \mu_0 I$$
$$B = \mu_0 (H + I) = 4\pi \times 10^{-7} (8000 + 0.184) T$$

④ Relative permeability (μ_r)

$$\mu_r = 1 + \chi = 1 + 2.3 \times 10^{-5}$$

Q A metal has permeability $\mu = 0.126$ T-m/A.

Find ① Relative permeability

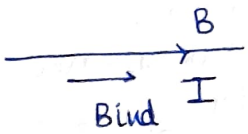
$$\mu_r = \frac{\mu}{\mu_0} = \frac{0.126}{4\pi \times 10^{-7}} = \frac{12.6 \times 10^5}{4 \times 3.14} = 100318$$

② Magnetic Susceptibility

$$\chi = \mu_r - 1 = 100317$$

Curie's law:

→ Given by Pierrie Curie



As temp ↑, Vibrations of atoms ↑, KE ↑, I ↓

$$\chi = \frac{I}{H}$$ If I ↓, χ ↓

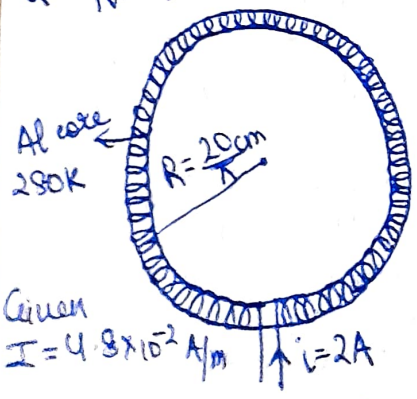
★ $\boxed{\chi = \frac{C}{T}}$ C = Curies Constant, T = Temp (Kelvin)

∴ $\boxed{\text{Temp} \uparrow, \chi \downarrow \quad \text{Temp} \downarrow, \chi \uparrow}$ $\boxed{\chi \propto \frac{1}{T}}$

→ Ferromagnetic substances have very high χ. If temperature of ferromagnetic substances is increased their χ will decrease and at a specific temp, they will behave like paramagnetic substance. This temp is known as Curie temp or Curie point (T_c).

∴ If $T > T_c$ $\boxed{\chi = \frac{C'}{T - T_c}}$

Q N = 400 turns



Find ① Magnetic intensity H

$$H = \frac{B_0}{\mu_0} = \frac{\mu_0 N i}{2\pi R \mu_0} = \frac{400 \times 2 \times 100}{2 \times \pi \times 20} = 2000 \text{ Amp/m}$$

② Magnetic Susceptibility at 280K

$$\chi = \frac{I}{H} = \frac{4.8 \times 10^{-2}}{2000} = 2.4 \times 10^{-5}$$

③ Magnetic Susceptibility at 320K

$$\chi \propto \frac{1}{T} \Rightarrow \frac{\chi_1}{\chi_2} = \frac{T_2}{T_1} \Rightarrow \frac{2.4 \times 10^{-5}}{\chi'} = \frac{320}{280}$$

$$\boxed{\chi' = 2.1 \times 10^{-5}}$$

④ Magnetization I at 320K.

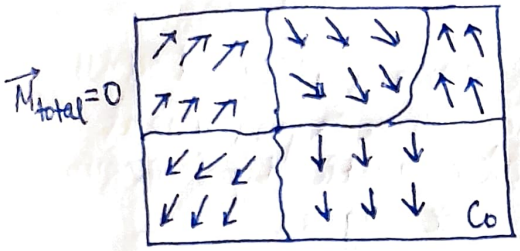
$I' = \chi' H = 2.1 \times 10^{-5} \times 2000 = 4.2 \times 10^{-2}$ A/m

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Ferromagnetism:

→ Ferromagnetic Substances ⇒ Fe, Ni, Co

→ Each atom has non-zero magnetic dipole moment. **(just like paramagnetic substances)**



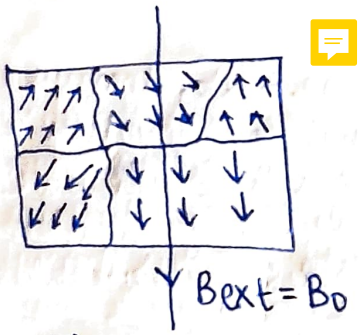
Due to a quantum phenomenon known as exchange coupling, nearby atoms interact and align along same direction.

These atomic groups are known as domain

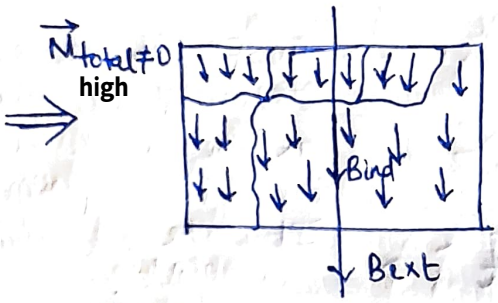
Size ≈ 1mm

No. of atoms ≈ 10^{11}

Now,



When $\vec{B}_{ext}$ is applied dipole moments in each domain starts aligning towards $\vec{B}_{ext}$.



Domain growth:

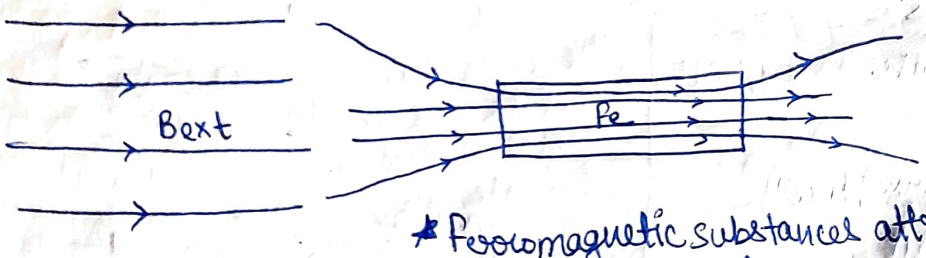
Domain along $\vec{B}_{ext}$ will grow in size and size of others will decrease.

① High M ⇒ High I ⇒ High B_{ind} ⇒ High B_{net}

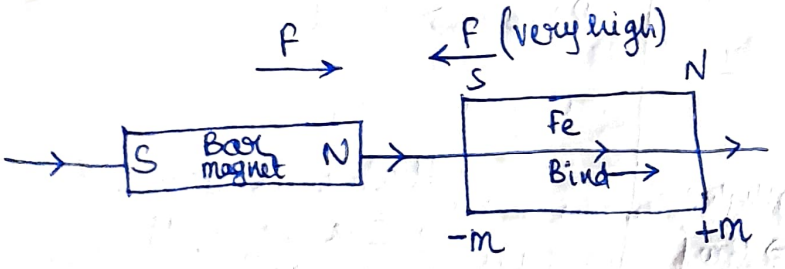
② $\chi = \frac{I}{H}$ ⇒ As I is high ⇒ χ is high (order of 1000)

③ $\mu_r = 1 + \chi$ ⇒ High μ_r (order of 1000)

④ High B as $B = \mu_r B_0$



★ Ferromagnetic substances attract external magnetic field.

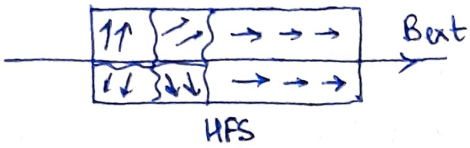


$B_{net} = B_{ext} + B_{ind}$

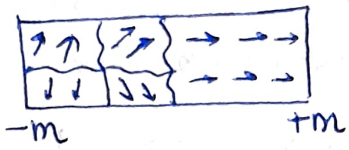
$B_{net} >> B_{ext}$

Hard Ferromagnetic Substances (HFS):

→ Examples are alloys : Al, Ni, Co, Cu



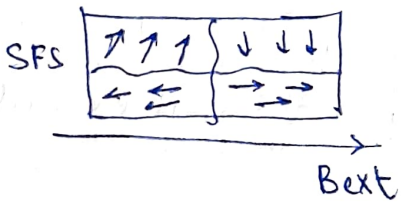
Now, we remove Bext



→ In HFS, if we remove Bext, domains will remain aligned rigidly as they were in presence of Bext and hence there is permanent magnetization of HFS and there is permanent magnetic dipole moment in it. Hence it behaves like permanent bar magnet.

Soft Ferromagnetic Substances (SFS):

→ Example: Soft iron

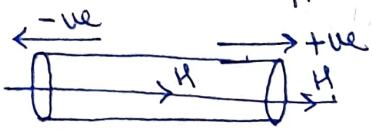


→ In SFS; if we remove Bext, domains will regain their original configuration ie as they were without Bext and hence there is no permanent magnetization. It does not behave like permanent bar magnet.

Hysteresis: in ferromagnetic substances

FMS $\xrightarrow{T_c}$ PMS

→ It is the phenomenon in which the value of a physical property lags behind the effect causing it.



Ferromagnetic Subs

① H=0 ⇒ I=0

② As H↑, I↑ (in +ve direction)

A → Saturation point
(Here I is maximum now)

③ As H↓, I↓

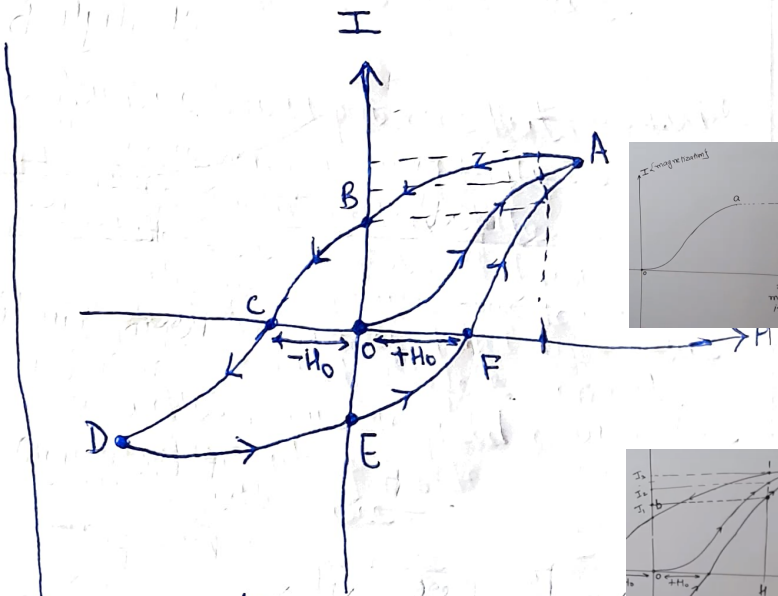
④ H=0, but I ≠ 0 as some domains will now be aligned as they were with H.

OB → Retentivity { (OB)HFS ⇒ High, (OB)SFS ⇒ low }

⑤ To demagnetise we further reduce H in -ve.

OC → Coercive force

↳ -ve magnetic intensity required to demagnetise ferro. subs.



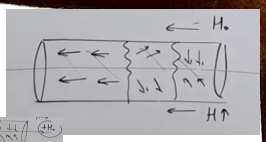
Hysteresis curve

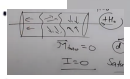
→ I is lagging H

→ Value of I depends on history of ferromagnetic subs.

This phenomenon is known as coercive

(5) Now we increase H in -ve and reach saturation point D.
(6) Again H=0, but I ≠ 0. OE ⇒ Retentivity = OB
(7) To demagnetise we increase H in +ve direcn. OF → Coercive force.





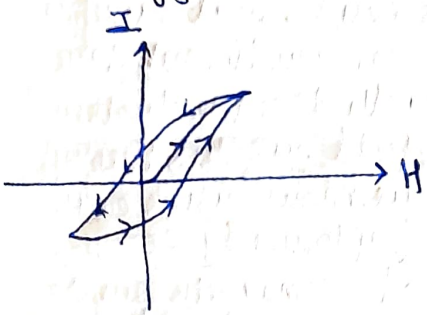
* Energy lost in 1 cycle ∝ area under the curve
of magnetisation-demagnetisation and remagnetisation

∴ High Retentivity – High Coercivity → More area ⇒ More energy loss

less Retentivity – less coercivity ⇒ less area ⇒ less energy loss

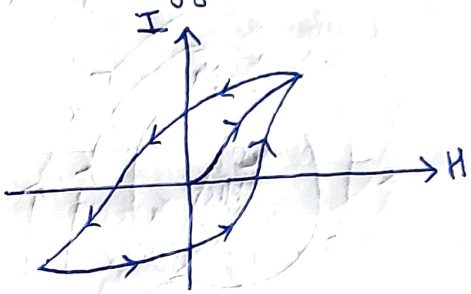
{ 1cycle ≡ Mag →
demag → Mag →
demag → Mag
ie, O → A → C → D
→ F → A }

Soft Ferromagnetic Substance:
→ less retentivity
→ less Coercive force
→ less area under curve
→ less energy loss



→ Useful in generators, transformers

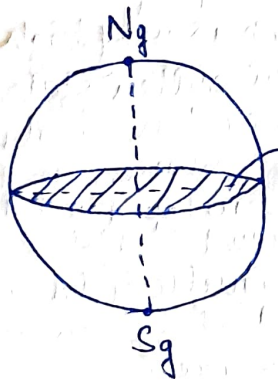
Hard Ferromagnetic Substance: ex. steel
→ more retentivity
→ more coercive force
→ more area under curve
→ more energy loss



→ Used to make permanent magnet

Earth's Magnetism:

Terrestrial magnetism:



Ng → Geographical North
Sg → Geographical South

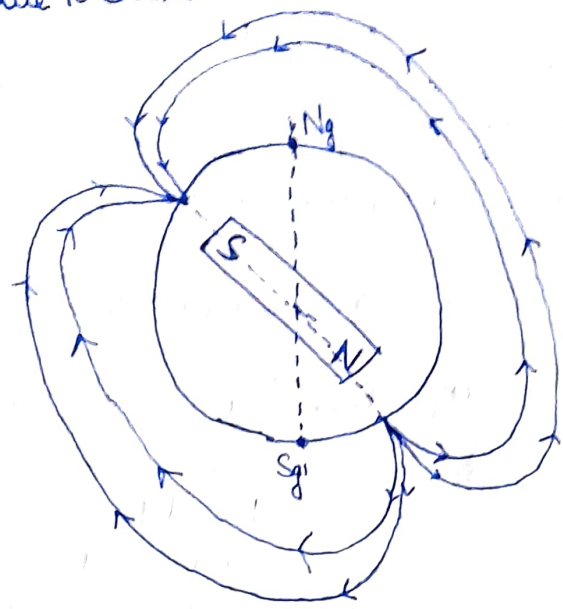
Geographical Equator

→ Geographical Meridian:
Geographical meridian at a pt. P is a vertical plane, containing pt. P and directed b/w geographical North pole and south pole.

→ Earth behaves like a large magnet.
→ B at earth's surface due to earth; B = 25μT – 65μT

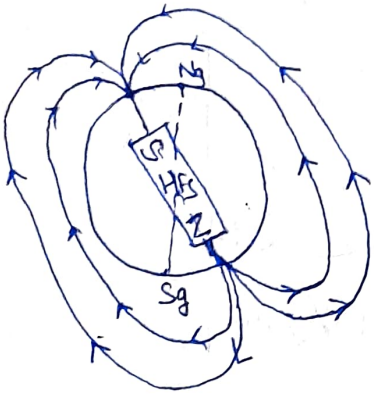
* 1 Gauss = 10^{-4} T

Magnetic Field lines due to earth:



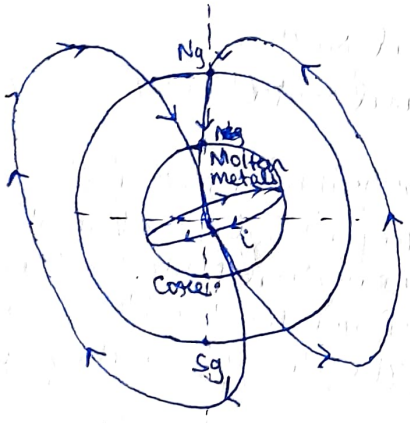
Reason behind magnetic field due to earth:

①



→ First thought can be that there maybe some hard ferromagnetic substances deep inside earth due to which we have the magnetic field but this theory failed because as we move inside earth, temp. rises & upto nearly 2000K-3000K, at this temp., ferromagnetic substance converts to paramagnetic substances and thus cannot produce the required magnetic field as experienced.

②



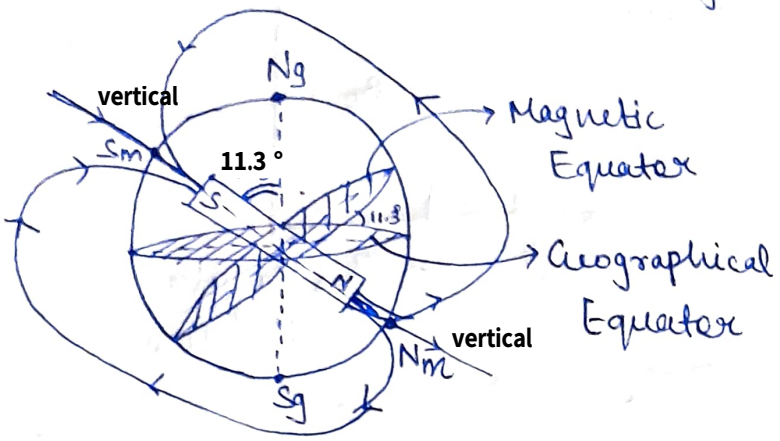
→ Another theory proposes that, as inside the core temp is very high so the metals are in molten state. As earth rotates, the molten metal ions also rotate in a plane due to which a current is developed and due to this we get a magnetic dipole moment. So earth's magnetic field may be due to movement of molten metal ions inside earth's core.
This theory is more acceptable.

Fact: (Not in syllabus) The plane in which the ion rotates also gets rotatated but it is very slow and takes lakhs of years for the polarity to get reversed.

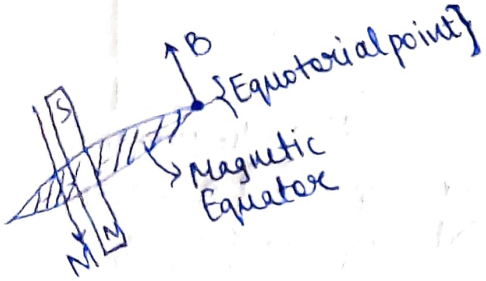
In around 7.6×10^7 years it is estimated that we had around 171 reversals.

Magnetic North and South Pole:

Nm → Magnetic North
Sm → Magnetic South



Q. If at magnetic equator $B = 3.1 \times 10^{-5}$ T then find magnetic dipole moment of earth. ($R = 6400$ km)



$$B = \frac{\mu_0}{4\pi} \frac{M}{R^3}$$

$$M = \frac{4\pi}{\mu_0} \cdot R^3 B$$

$$= 10^7 \times (6400000)^3 \times 3.1 \times 10^{-5}$$

Magnetic Meridian: It is a vertical plane containing a freely suspended bar magnet.

OR

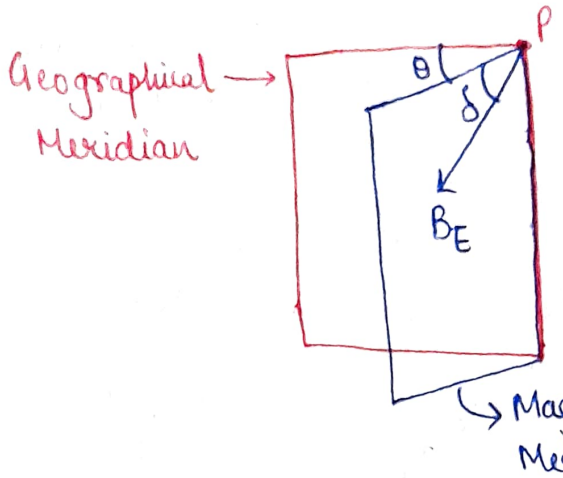
It is a vertical plane containing earth's magnetic field at that point.

Angle of declination (θ): Angle of declination at a point is angle b/w geographical meridian and magnetic meridian.

Angle of dip / Angle of inclination (δ): Angle between horizontal and freely suspended bar magnet in magnetic meridian is known as angle of dip (δ).

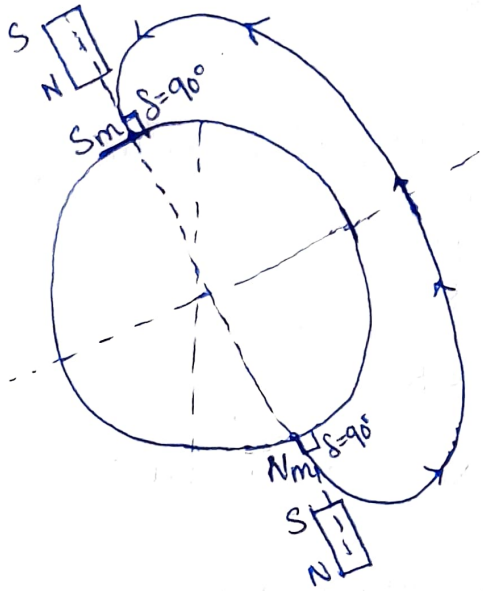
OR

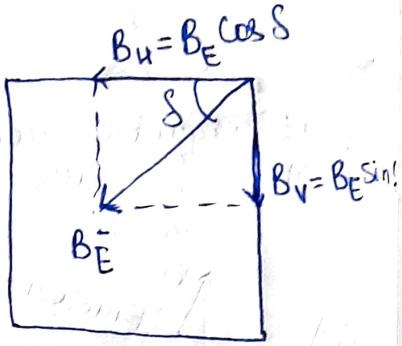
Angle between horizontal and earth's magnetic field in magnetic meridian is known as δ.



θ → Angle of declination

δ → Angle of dip





Magnetic Meridian

Hor^z component of $B_E = B_H = B_E \cos\delta$

Vert. component of $B_E = B_V = B_E \sin\delta$

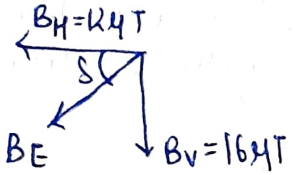
$$B_E = \sqrt{B_H^2 + B_V^2}$$

$$\tan\delta = \frac{B_V}{B_H}$$

→ At magnetic Equator ⇒ $\delta = 0$ $(B_E = B_H, B_V = 0)$

→ At Sm and Nm ⇒ $\delta = 90°$ $(B_E = B_V, B_H = 0)$

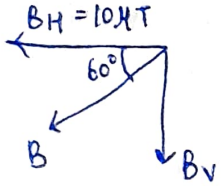
Q Given $B_H = 12\mu T$ and $B_V = 16\mu T$. Find ① Angle of dip ② Earths magnetic field.



① $\tan\delta = \frac{B_V}{B_H} = \frac{16}{12} = \frac{4}{3} \Rightarrow \delta = 53°$

② $B_E = \sqrt{16^2 + 12^2} = \sqrt{400} = 20\mu T$

Q Given $B_H = 10\mu T$ and $\delta = 60°$. Find ① B_V ② B



① $\tan 60° = \frac{B_V}{10} \Rightarrow B_V = 10 \tan 60 = 10\sqrt{3}\ \mu T$

② $B_H = B\cos 60° \Rightarrow B = \frac{B_H}{\cos 60°} = \frac{10}{1/2} = 20\mu T$